LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Daily Practice: Ch 13 Our Environment

Chapter / Topic: Our Environment · Week 4 of the 90-day plan

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DAY 1: ECOSYSTEMS AND FOOD CHAINS

Q1. Which of these is an abiotic component of an ecosystem?

Answer: (a) temperature (rainfall, soil and minerals are abiotic too)

Q2. An animal that eats both plants and animals is called:

Answer: (d) an omnivore

Q3. In the food chain plants → insects → frogs → snakes, the frog is a:

Answer: (a) secondary consumer (third trophic level)

Q4. Arrange hawk, grass, snake and rat into a food chain. Which trophic level is the snake at?

Answer: Grass → rat → snake → hawk. The snake is at the third trophic level (secondary consumer).

Q5. Give one example each of a land food chain and a water food chain, with three trophic levels.

Answer: Land: grass → deer → tiger (or grass → goat → human).
Water: phytoplankton → zooplankton → fish (any correct examples).

Q6. In a pond, you find water plants, tadpoles that eat the plants, small fish that eat the tadpoles, and bacteria on the dead remains at the bottom.…

Answer: Producer: water plants. Primary consumer: tadpoles. Secondary consumer: small fish. Decomposer: bacteria.

DAY 2: ENERGY FLOW AND NUTRIENT CYCLING

Q7. At each trophic level, most of the energy taken in is:

Answer: (b) used for life processes and lost as heat; only about 10% is passed on

Q8. In the food chain grass → deer → lion, the grass has 20,000 J of energy. About how much reaches the lion?

Answer: (b) 200 J: deer get 2,000 J (10%) and lions 200 J (10% of that)

Q9. Energy enters an ecosystem through the:

Answer: (d) producers, which capture sunlight in photosynthesis

Q10. Why does the food chain grass → human make more energy available to humans than grass → goat → human?

Answer: The first chain has fewer trophic levels. At every step about 90% of the energy is lost, so with one step fewer, much more of the energy trapped by the grass reaches humans.

Q11. In the food chain phytoplankton → zooplankton → fish → kingfisher, the kingfisher receives 2 J of energy. How much energy did the phytoplankton trap?…

Answer: Work backwards, multiplying by 10 at each step: kingfisher 2 J ← fish 20 J ← zooplankton 200 J ← phytoplankton 2,000 J.

Q12. Energy flows through an ecosystem in one direction, but nutrients go round in a cycle. Explain the difference.

Answer: Energy from the sun passes from one trophic level to the next and is lost as heat at every step; it never comes back to the plants. Nutrients (minerals) in dead plants and animals are broken down by decomposers and returned to the soil, where plants take them up and use them again.

DAY 3: OZONE, WASTE AND CHALLENGE

Q13. **Assertion (A):** At ground level, ozone is a deadly poison. **Reason (R):** An ozone molecule is made up of two oxygen atoms.

Answer: (c) A is true, but R is false: ozone (O_3) has three oxygen atoms.

Q14. In 1987, an international agreement led by the United Nations Environment Programme froze the production of:

Answer: (c) CFCs, at their 1986 levels, to protect the ozone layer

Q15. Which of these will take the longest to break down in soil?

Answer: (d) a polythene bag: it is non-biodegradable

Q16. A school's eco-club weighed the canteen waste for one week: 40 kg of leftover food and peels, 12 kg of paper plates and napkins, 8 kg of plastic…

(i) Leftover food and peels, and paper plates and napkins.
(ii) Total = 40 + 12 + 8 + 2 = 62 kg. Biodegradable = 40 + 12 = 52 kg. 52/62 × 100 ≈ 84%.
(iii) Food waste: compost it, and use the compost in the school garden. Plastic: use steel plates and glasses that can be washed and reused, and send bottles for recycling.

Q17. Why do we never need to clean ponds or lakes, while an aquarium must be cleaned regularly?

Answer: A pond is a natural, balanced ecosystem: its decomposers break down wastes and dead remains and return the nutrients. An aquarium is a small, artificial ecosystem with few decomposers, so wastes build up and must be removed by cleaning.